



Outline of the lecture

The history of Informatics & HCI

Our roots and evolution

Conceptual Foundations

Information Processing | Gestalt Theory | Functional Approaches

Action Research & Praxeology

Work(place) studies & ethnography | Contextual Approaches |
Praxeology & Appropriation | Activity Theory

Structuration & Infrastructuring

A holistic look at Human-System-Interaction

HCI in Practice

Designing | Prototyping | Evaluation

Users and Use Context

Basics

Primary Users	Secondary Users	Tertiary Users
Those who directly interact with the technology	Those who use it via an intermediary; may have an influence about primary users (using/not using it)	Those who are affected by the technology use and decide its purchase/implementation

Use Context

- Where are users located?
- What are the social, cultural and organisational aspects of the context?
- With what/whom do they interact?
- What role do technologies play in the context?
- How do these aspects impact upon the way that users do things?

stakeholders

motivations, and goals

understand people's behaviors

An individual or group with an interest in the success of a product/organization in delivering intended results and maintaining the viability of the organization's products and services. **Stakeholders influence** programs, products, and services (Kotonya and Sommerville, 1998)

Before Use Context

Problems and deficits in HCI in early 1980

Problem situation

HCI concentrates on users and interface. However, interaction with other people, context, culture, groups is rather neglected (Norman, 1980).

Observable deficits in cognition-based HCI*

1. Early advanced user interfaces assumed the **users mental models or generic users**, their cognition - without concern for qualifications, work environment, etc.
2. The **role of the artifact** as it stands between the user and the materials, objects and outcomes was ill understood.
3. In validating findings and designs there was a heavy focus on novice users whereas everyday use by experienced users and concerns for the development of **expertise were hardly addressed**.
4. Detailed task analysis **failed to capture the complexity of real-life action**
5. From the point of view of complex work settings, it was striking how most **HCI focused on one user – one computer** in contrast to the ever-ongoing cooperation and coordination of real work.
6. Users were mainly seen **as objects of study**.

*Bertelsen, Olav W., and Susanne Bødker. "Activity theory." *HCI models, theories, and frameworks: Toward a multidisciplinary science* (2003): 291-324.

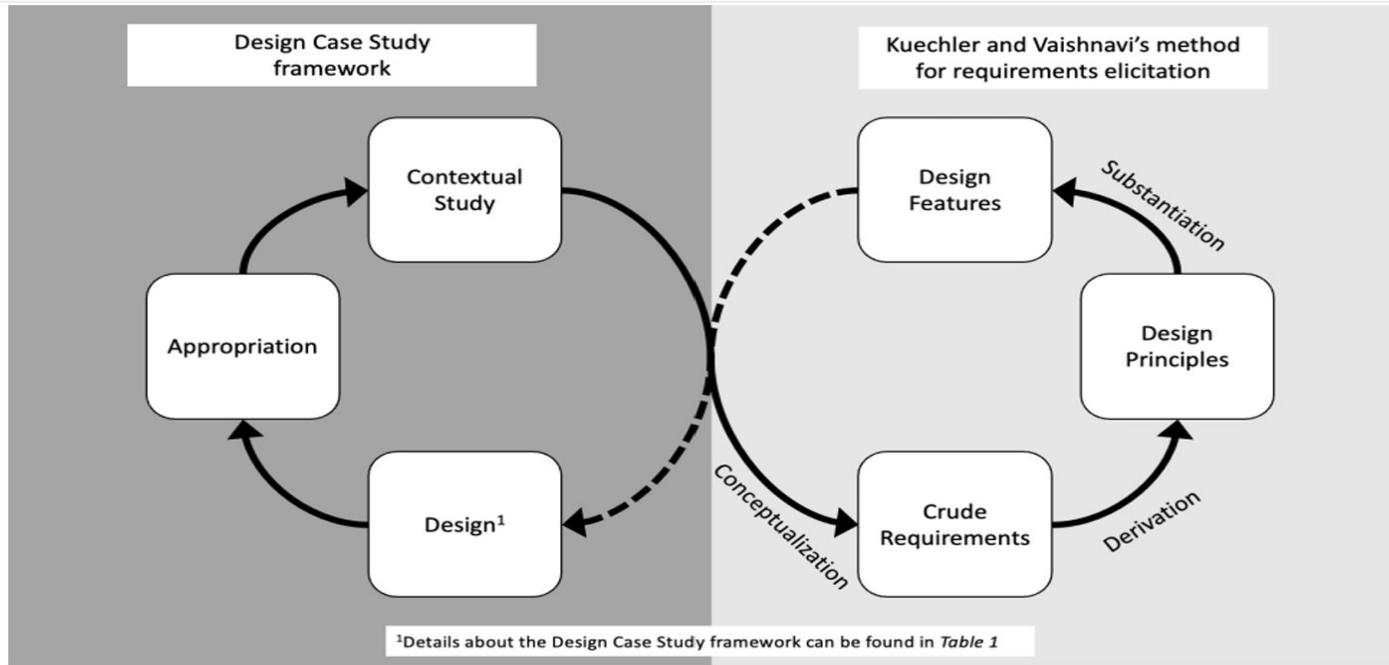


Figure 4. Blending adaptation of Kuechler and Vaishnavi's for requirements elicitation within stages of DCS.

Pragmatic Approach

Practice Centered Design / Design Case Studies

Formative and Contextual Approaches

Not a clear difference

	Formative	Contextual
Purpose	Shape and improve design	Understand real-world context
Stage	During design (iterative)	Before or during early design
Outcome	Inform design, Design insights and refinements	Contextual understanding and user requirements

Contextual Approaches

Basics

Approach	Description	Type of Data Collected
Contextual Inquiry	A semi-structured interview conducted in the user's real work or use environment; the researcher observes and asks questions as the user performs tasks.	Task flows, workarounds, artifacts used, implicit practices
Ethnographic Fieldwork	Long-term observation of users in their natural environment; researcher becomes part of the community or workflow.	Social interactions, routines, culture, communication patterns
Participant Observation	Researcher participates in the user's activity while observing interactions.	Detailed, experiential understanding of behaviors and meanings
Diary Studies / Experience Sampling	Participants record activities, feelings, or interactions with a system over time.	Longitudinal data on user experiences, habits, frustrations
Cultural Probes	Creative, open-ended prompts (e.g., postcards, disposable cameras, diaries) given to users to express their experiences.	Subjective and emotional insights into user values and contexts
Shadowing / Field Studies	Researcher shadows a participant during tasks to observe use in context.	Workflow and contextual factors (e.g., interruptions, collaboration)

Formative and Contextual Approaches

Basics

Formative Method	Description / Purpose	Type of Data	When Used in Design Process
Interviews (Semi-structured or Unstructured)	Explore users' experiences, needs, motivations, and pain points in their own words.	Qualitative	Context study
Focus Groups	Group discussion to explore attitudes, preferences, or reactions to ideas or prototypes.	Qualitative	Context to mid design
Participatory Design Workshops	Involve users directly in co-designing solutions or interfaces.	Qualitative / Design artifacts	Mid design
Card Sorting	Explore how users group and label information for navigation or categorization tasks.	Qualitative / Quantitative	Early or mid design
Storyboarding / Scenario-Based Design	Develop stories or visual scenarios of how users might use a system.	Qualitative / Design artifact	Early or mid design
Prototyping (Low- to Mid-Fidelity)	Create simplified models of the design to test ideas early.	Qualitative / Mixed	Mid design (iterative)
Heuristic Walkthroughs (Formative Usability Evaluation)	Designers or experts identify usability issues early, before full testing.	Qualitative	Mid design
Field Studies / Field Trials	Observe how users interact with prototypes in real settings.	Qualitative / Mixed	Mid to late design

Intensifying the Problem: Work(place) studies

Basic idea: HCI takes place against the background of a work situation (everyday practice).

- ▶ **Happening in the wild (cannot be Lab Studies)**

Cognitive approaches fail to capture the complexities and nuances of real-world environments where work actually happens - involving unexpected events, evolving goals, and constant adaptation - so interruptions, social interactions, environmental constraints cannot be adequately modeled

- ▶ **Is Messy**

Real work is messy, non linear and collaborative - requiring iterative, overlapping, and occasionally conflicting actions

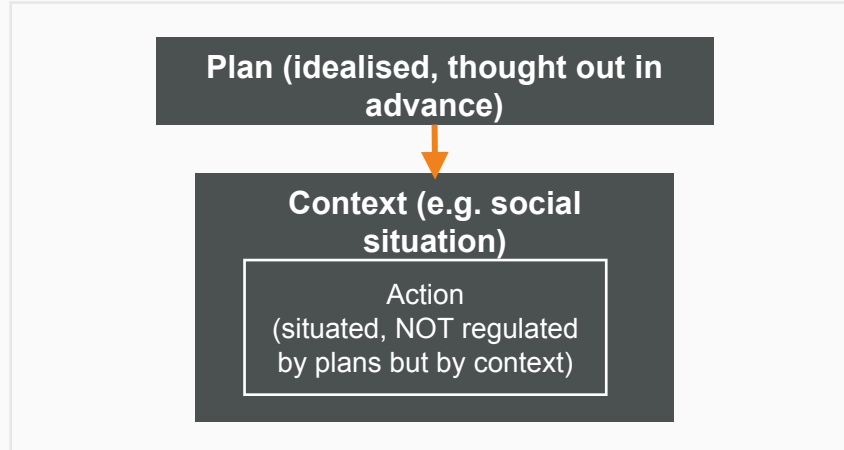
- ▶ **Turn to practice**

By focusing on actual practices, researchers can uncover tacit, context-specific knowledge that is invisible in purely cognitive approaches, which prioritize universal theories over situated understanding



Plans & Situated Action

Ethnographical foundation for Work(place) studies: a subjective orientation to action



Pioneer: Lucy Suchman, 1987

Planning and representation, central to symbolic theories, are claimed to be irrelevant in everyday human activity (Situated action: A symbolic interpretation by Vera and Simon, 1993)



Cognitive Models (esp. from psychology)

People's behaviour driven by intellectual models, purpose, aim, motivation etc. Explained by theories.



Situated Action

Humans and their interactions with the world cannot be understood using symbol-system models and methodology, but only by observing them within real-world contexts or building nonsymbolic models of them.



Ethnographical approach

Practical orientation: "What is happening here?" vs. "which theories correspond to what is happening?".

Origins lie in the research of indigenous peoples "from the inside" –participation in the field is an integral part of this method! Its principal direction is ethnomethodology: "*We cannot recognize one objective, actual practice but rather the actor's intended practice*".

"The world is changing rapidly, with new views on the role of design in the world as well as a growing realization that the existing model of design practice had its historical origins in the industrialized nations of the world as a service to industry. Today's designers have moved far beyond these historical origins. One result is greater awareness of the deterioration of the economies, environment, and culture in what has been called the "colonization of the world."

Meyer, M. W., & Norman, D. (2020). Changing Design Education for the 21st Century. *She Ji: The Journal of Design, Economics, and Innovation*, 6(1), 13-49. Changing Design Education for the 21st Century - Science Direct United Nations. (2019). (Source: Science Direct)

Role of a Designer
